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01/16/26

HW 0

per tsp
↓

2. $\frac{3}{4}$ tsp NaCl = 6g NaCl = 4.5g NaCl

1 lbm peanut $\rightarrow$ 16 oz peanut

16 oz $\cdot$ 5mg NaCl $\rightarrow$ 80 mg NaCl

~~Total salt in recipe = 4.5g~~

salt

NaCl is 39.34% Na $\rightarrow$ 1.77g Na in added

Total sodium 1.85g

a) 16 oz light salt peanut

16 $\cdot$ 75 mg $\rightarrow$ 1,200 mg Na $\rightarrow$ 1.2g Na

1.85 - 1.2 = 0.65g Na added $\div$ 0.3934

1.65g NaCl $\div$ 6 g/tsp = 0.275 tsp salt

b) 16 $\cdot$ 150 mg $\rightarrow$ 2,400 mg Na $\rightarrow$ 2.4g Na

Already too much sodium. - 0.55g Na added

$\div$ 0.3934 = -1.41 $\div$ 6 $\rightarrow$ 0.233 tsp of "anti salt"

c) Lightly salted peanuts are OK to use for PB,
but regularly salted peanuts are too salty

Pr. 1: 20/20

2)

200 g CuCl_x Mixture 100 g Cu initial

CuCl_2 is 47.3% Cu

CuCl is 64.2% Cu

no balanced eqns, -1 pt

pc. 2: 15/20

$$\rightarrow 15.98\% \text{ CuCl} \quad 84.02\% \text{ CuCl}_2$$

$$X(0.642) + (1-X)(0.473) = 0.5 \quad (\text{ratio of metallic Cu to mixture})$$

$$0.169X = 0.027$$

$$X = 0.1598$$

$$0.1598 \text{ CuCl} \quad 0.8402 \text{ CuCl}_2$$

-4 pts

3) $C_p(300) = 28.11 - 3.68 \cdot 10^{-6} \cdot 300 + 1.746 \cdot 10^{-5} \cdot 300^2 -$

a) $1.065 \cdot 10^{-8} \cdot 300^3$

$C_p(300) = 29.39 \text{ J/mol}$

$C_p(700) = 28.11 - 3.68 \cdot 10^{-6} \cdot 700 + 1.746 \cdot 10^{-5} \cdot 700^2 - 1.065 \cdot 10^{-8} \cdot 700^3$

$C_p(700) = 33.01 \text{ J/mol}$

difference = + 3.62 J/mol

$C_p(t) = 28.11 - 3.68 \cdot 10^{-6} \cdot t + 1.746 \cdot 10^{-5} \cdot t^2 - 1.065 \cdot 10^{-8} \cdot t^3$

$C_{pt}(0.3) = 29.39, C_{pt}(0.7) = 33.01, \text{ difference} = 3.62 \text{ J/mol}$

no part b, -10 pts

2/20

??? No give what this stuff is (1)
 For $R(h) \leq 101 \forall h \in R$

$$4. \int_1^h \frac{1}{40-4\sqrt{x-1}} dx$$

$$\downarrow$$

$$\frac{1}{4} \int_1^h \frac{1}{10-\sqrt{x-1}} dx$$

$$\frac{1}{4} \int_0^{\sqrt{h-1}} \frac{2u}{10-u} du$$

Set $u = \sqrt{x-1}$

thus $x = u^2 + 1$

$dx = 2u du$

if $x=1, u=0$

$x=h, u=\sqrt{h-1}$

Save for later

$$\frac{1}{2} \int_0^{\sqrt{h-1}} \frac{u}{10-u} du \quad \text{ugly}$$

$$\frac{u}{10-u} = -1 + \frac{10}{10-u}$$

$$\downarrow$$

$$\frac{1}{2} \int_0^{\sqrt{h-1}} \left(-1 + \frac{10}{10-u} \right) du$$

$$\downarrow$$

$$-u + 10 \ln|10-u| \Big|_0^{\sqrt{h-1}}$$

$$\frac{1}{2} \left(-\sqrt{h-1} + 10 \ln|10-\sqrt{h-1}| + 10 \ln 10 \right)$$

$$\downarrow$$

$$-\frac{\sqrt{h-1}}{2} + 5 \ln|10-\sqrt{h-1}| + 5 \ln 10$$

Pr. 4: 20/20

5) $7 \text{ kcal/g} \rightarrow 7 \text{ kJ/mol}$

$2 \text{ kcal} = 4.184 \text{ kJ}, 29.288 \text{ kJ/g}$

EtOH Molar Mass = 46.069 g/mol

$29.288 \cdot 46.069 \rightarrow 1349.269 \text{ kJ/mol.}$

Compared to liquid phase EtOH combustion, this value is very similar (albeit positive). This does make sense since the primary method of EtOH consumption (phase-wise) is liquid EtOH. Combustion rxns always* have negative enthalpies, as combustion reactions release heat, which is why they are used to power things.

* Never seen otherwise.

Pr. 5: 20/20